

Algebraic invariants associated to graph coloring

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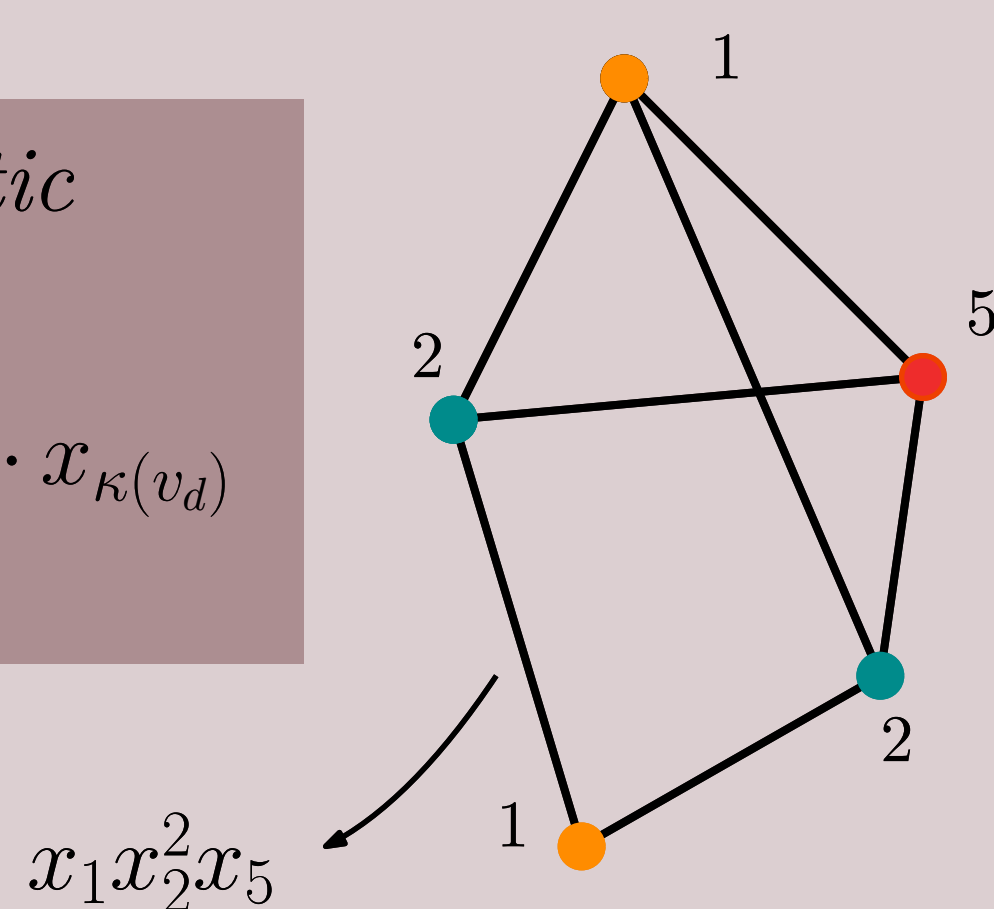
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CLASSICAL VS. SIGNED GRAPH COLORING

Def: Given a simple graph $G = (V, E)$, a proper coloring of G is a map $\kappa : V \rightarrow \mathbb{N}^*$ verifying $\kappa(u) \neq \kappa(v)$ if $\{u, v\} \in E$

In [Sta95] Richard P. Stanley defined the *chromatic symmetric function* of G as:

$$X_G(x_1, x_2, \dots) := \sum_{\kappa \text{ proper}} x^\kappa = \sum_{\kappa \text{ proper}} x_{\kappa(v_1)} x_{\kappa(v_2)} \cdots x_{\kappa(v_d)}$$



$$x_1 x_2^2 x_5$$

$$X_G \in \text{Sym} = \{f \in \mathbb{Q}[[x_1, x_2, \dots]] : \sigma * f = f \quad \forall \sigma \in \mathfrak{S}\}$$

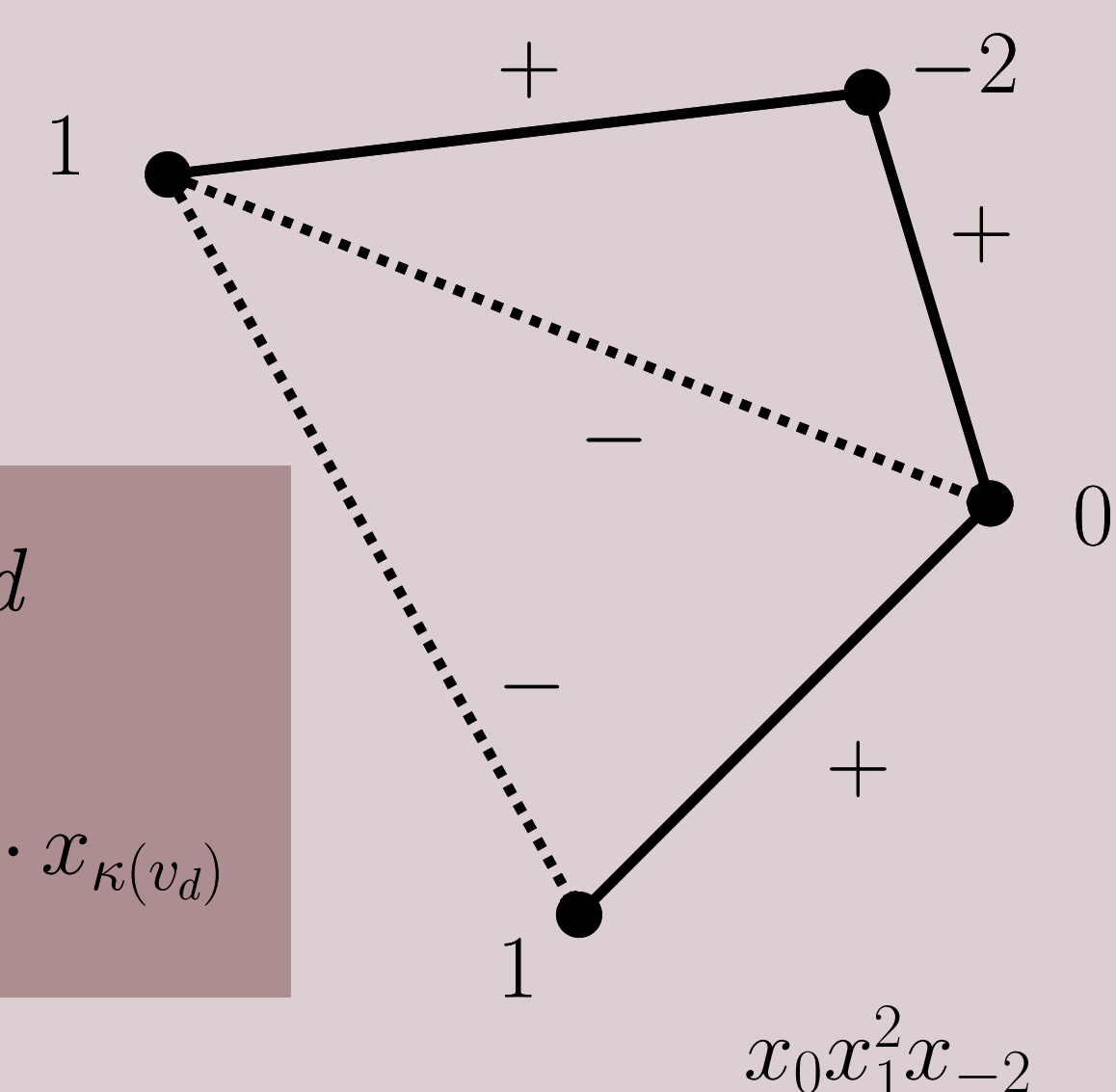
$$\text{where } \sigma * f(x_1, x_2, \dots) := f(x_{\sigma(1)}, x_{\sigma(2)}, \dots).$$

Def: A proper coloring of a signed graph $\Sigma = (V, E^+, E^-)$ is a map $\kappa : V \rightarrow \mathbb{Z}$ verifying:

- $\kappa(u) \neq \kappa(v)$ if $\{u, v\} \in E^+$
- $\kappa(u) \neq -\kappa(v)$ if $\{u, v\} \in E^-$

In [KT21] they define the *chromatic signed symmetric function* of Σ as:

$$X_\Sigma(\dots, x_{-1}, x_0, x_1, \dots) := \sum_{\kappa \text{ proper}} x_{\kappa(v_1)} x_{\kappa(v_2)} \cdots x_{\kappa(v_d)}$$



$$x_0 x_1^2 x_{-2}$$

$X_\Sigma \in \text{SSym}$, the algebra of signed symmetric functions

$$\text{SSym} := \{f \in \mathbb{Q}[[\dots, x_{-1}, x_0, x_1, \dots]] : \sigma * f = f \quad \forall \sigma \in S\mathfrak{S}\}$$

where $S\mathfrak{S}$ are the permutations of $\mathbb{Z}$ verifying $\sigma(i) = -\sigma(-i)$ for all $i \in \mathbb{Z}$.

THE GEOMETRIC POINT OF VIEW

Given a graph $G = (V, E)$, we can associate it an hyperplane arrangement $\mathcal{H}_G$ in $\mathbb{R}^{|V|}$ as:

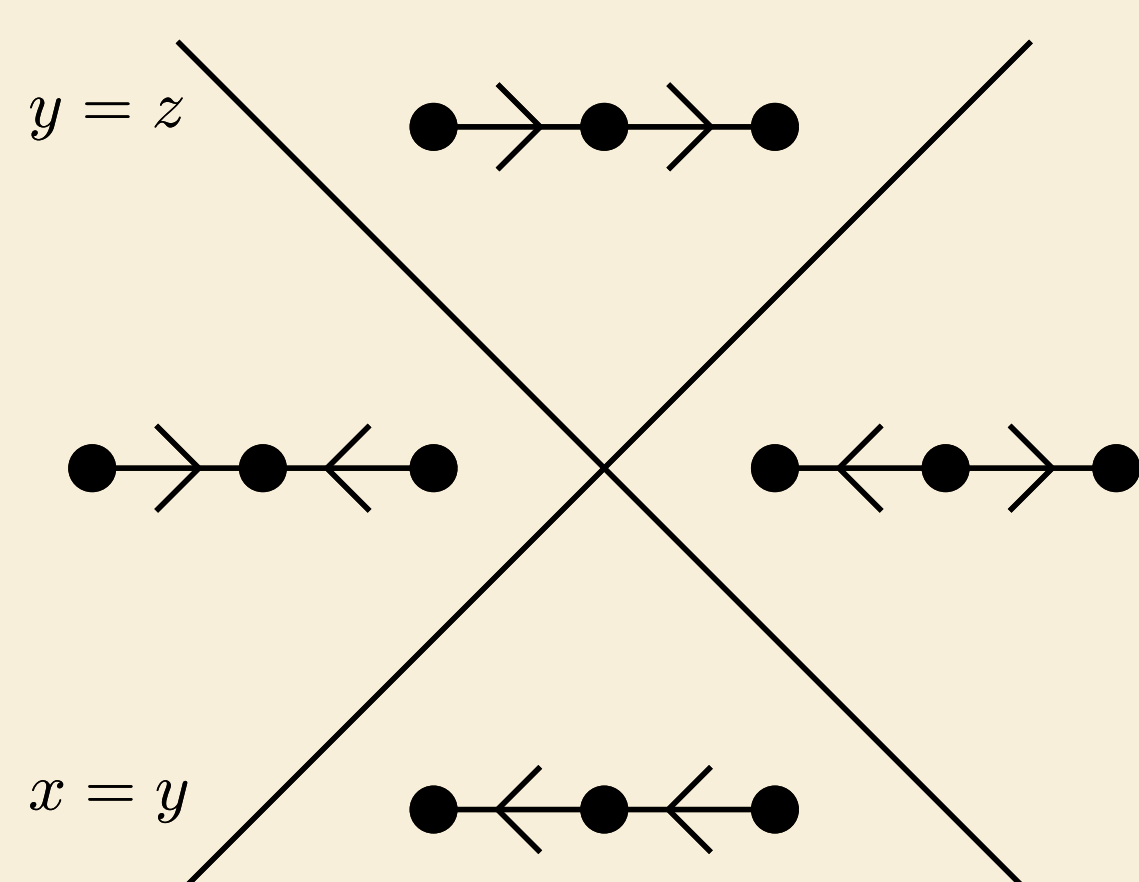
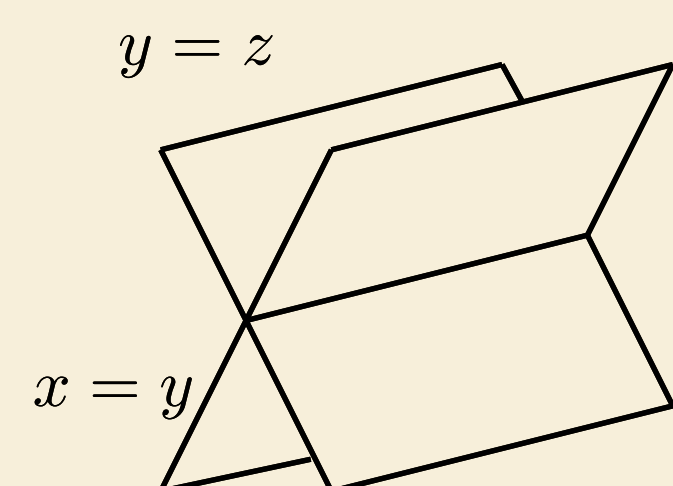
$$\mathcal{H}_G := \{\{x_i = x_j\} : \{i, j\} \in E\}$$

Then we have the bijections:

$$\left\{ \begin{array}{l} \text{Proper colorings of } G \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Integer points} \\ \text{in } (\mathbb{R}^+)^{|V|} \setminus \mathcal{H}_G \end{array} \right\}$$

$$\left\{ \begin{array}{l} \text{Acyclic orientations} \\ \text{of } G \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Chambers of} \\ (\mathbb{R}^+)^{|V|} \setminus \mathcal{H}_G \end{array} \right\}$$

Example : $x \quad y \quad z$



$$\text{And then } X_G(x) = \sum_{C \text{ chamber}} \left(\sum_{\alpha \in C \cap \mathbb{N}^{|V|}} x^\alpha \right)$$

Given a signed graph $\Sigma = (V, E^+, E^-)$, we can associate it an hyperplane arrangement $\mathcal{H}_\Sigma$ in $\mathbb{R}^{|V|}$ as:

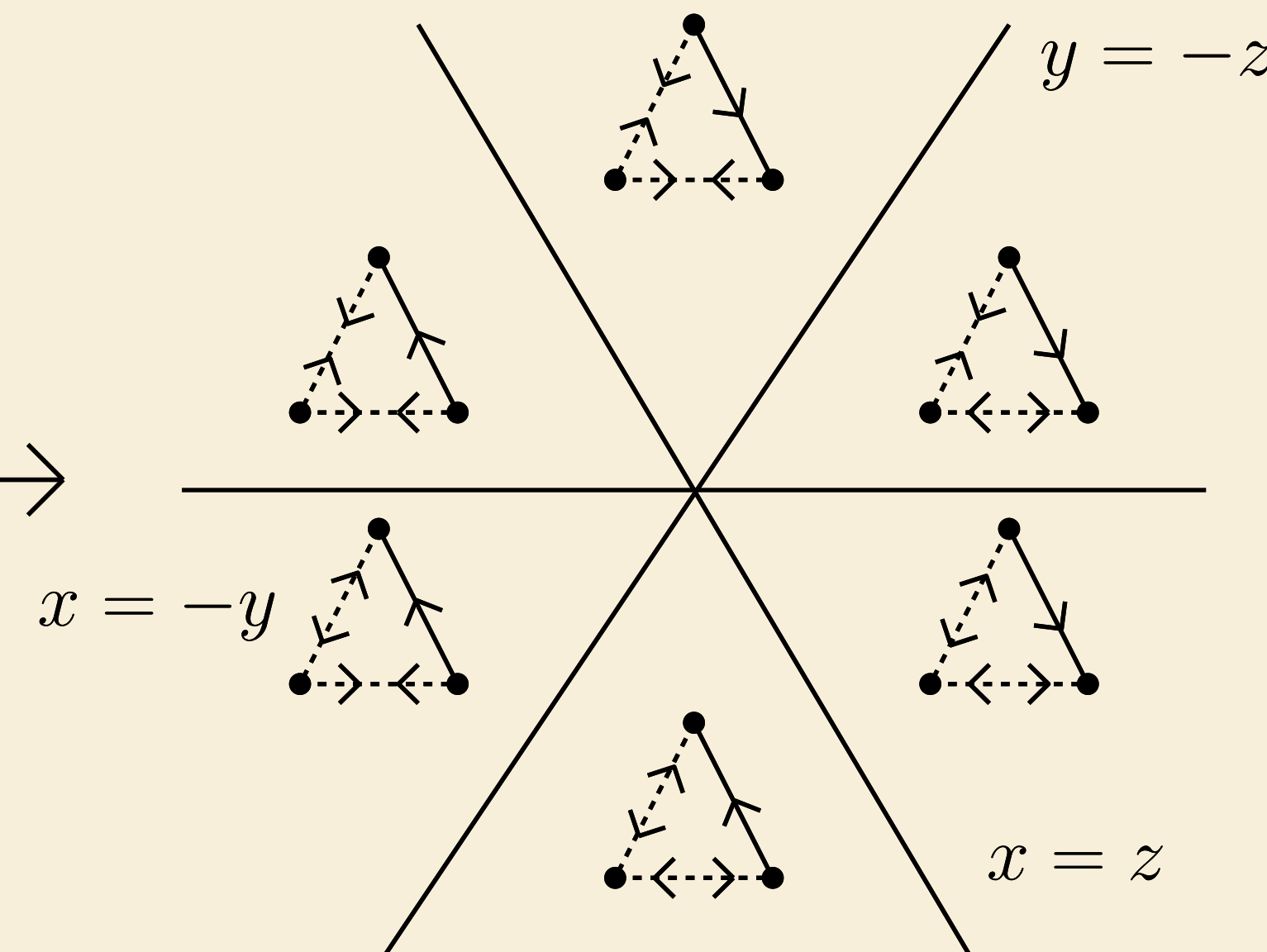
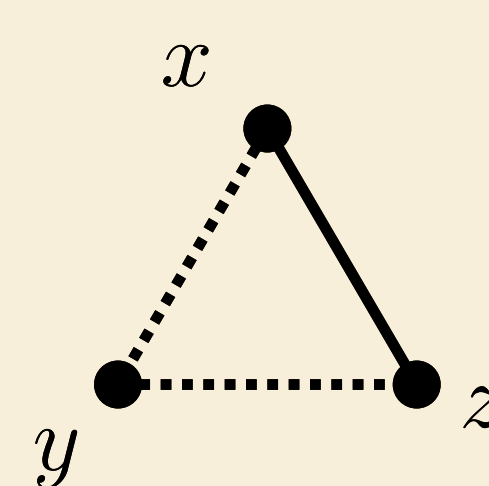
$$\mathcal{H}_\Sigma := \{\{x_i = x_j\} : \{i, j\} \in E^+\} \cup \{\{x_i = -x_j\} : \{i, j\} \in E^-\}$$

Then we still have the bijections:

$$\left\{ \begin{array}{l} \text{Proper colorings of } \Sigma \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Integer points} \\ \text{in } \mathbb{R}^{|V|} \setminus \mathcal{H}_\Sigma \end{array} \right\}$$

$$\left\{ \begin{array}{l} \text{Acyclic orientations}^* \\ \text{of } \Sigma \end{array} \right\} \xrightarrow{[Z91]} \left\{ \begin{array}{l} \text{Chambers of} \\ (\mathbb{R}^+)^{|V|} \setminus \mathcal{H}_G \end{array} \right\}$$

Example :



THE QUASISYMMETRIC INVARIANT

In [SW14], [E17] given a directed graph $\vec{G}$ and a proper coloring of it κ they define a statistic $\text{asc}_{\vec{G}}(\kappa)$ **that only depends on the chamber of κ** and they define the *chromatic quasisymmetric function* of $\vec{G}$ as

$$X_{\vec{G}}(x_1, x_2, \dots; t) = \sum_{\kappa \text{ proper}} t^{\text{asc}(\kappa)} x^\kappa \in \text{QSym}$$

BIBLIOGRAPHY

[E17] Ellzey (2017). *A directed graph generalization of chromatic quasisymmetric functions*, arXiv preprint

[K21] Kuroda & Tsujie (2021). *Chromatic signed-symmetric functions of signed graphs*, arXiv preprint

[SW16] Shareshian & Wachs (2016). *Chromatic quasisymmetric functions*, Adv. Math.

[S95] Stanley (1995). *A symmetric function generalization of the chromatic polynomial of a graph*, Adv. Math.

[Z91] (1991). *Orientation of signed graphs*, Eur. J. Comb.

A SIGNED QUASISYMMETRIC INVARIANT

In an analogous way, given a signed directed graph $\vec{\Sigma}$ and a proper coloring κ , we define an statistic $\text{asc}_{\vec{\Sigma}}(\kappa)$ and we define the *signed chromatic quasisymmetric function* of $\vec{\Sigma}$ as

$$X_{\vec{\Sigma}}(\dots, x_{-1}, x_0, x_1, \dots; t) = \sum_{\kappa \text{ proper}} t^{\text{asc}(\kappa)} x^\kappa$$

and we have called $SQSym$ the space where the coefficients of each power of t live (i.e. $X_{\vec{\Sigma}}(x; t) \in SQSym[t]$).

RESULTS AND FORTHCOMING WORK

- Find different basis for $SQSym$ and express $X_{\vec{\Sigma}}$ in them
- Study the Hopf algebra structure of $SQSym$
- For which $\vec{\Sigma}$ is $X_{\vec{\Sigma}}$ signed symmetric?
- Attack the problem with combinatorial geometry techniques